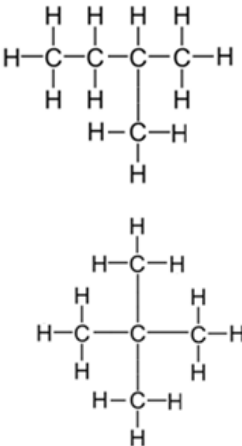


Higher Tier only questions

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			award (1) for any of following - the longer the chains, the higher the boiling point - the more carbon atoms, the higher the boiling point - the larger the molecules, the higher the boiling point - the larger M_r, the higher the boiling point the higher the boiling point, the lower in the column the fraction is collected (1) accept converse argument throughout e.g. the shorter the chains, the lower the boiling point etc. if no reference to boiling point award (1) for 'the bigger the molecule, the lower down the column it collects'	2			2		
	(b)	(i)		C_4H_8		1		1	1	
		(ii)		any two of following high temperature / heat strongly - catalyst - absence of air ignore any reference to pressure – neutral answer	1			1		
		(iii)		any of following - small(er) fractions are more useful / used as fuels - more demand for small(er) fractions - produces alkenes / unsaturated molecules - conserves crude oil supplies - produces monomers to make plastics	1			1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		same molecular formula but different structural formulas	1			1		
		(ii)		award (1) for each correct isomer 	2			2		
	(d)			1.2 g of hydrogen (1) $C = \frac{7.2}{12}/0.6$ and $H = \frac{1.2}{1}/1.2$ (1) 1:2 ratio / alkenes have the general formula C_nH_{2n} / alkenes always have double the hydrogen to carbon (1) third mark to be awarded only if the correct ratio is found		2	1	3	2	
				Question 3 total	7	3	1	11	3	0